



Quiz
Test yourself 5

TEST YOURSELF 5

ANSWERS ON P. 578

5.01

1 Use the commutative and associative laws to evaluate each expression.

- a** $7 + 18 + 3$ **b** $91 + 18 + 9$ **c** $2 \times 17 \times 5$
d $25 \times 31 \times 4$ **e** 13×30 **f** 7×400

5.02

2 Use the distributive law to evaluate each expression.

- a** 22×9 **b** 17×8 **c** 27×11 **d** 13×12

5.03

3 Examine this number pattern, then write the general rule using a variable.

$$3 - (-3) = 2 \times 3$$

$$5 - (-5) = 2 \times 5$$

$$11 - (-11) = 2 \times 11$$

$$2 - (-2) = 2 \times 2$$

$$8 - (-8) = 2 \times 8$$

5.03

4 Simplify each expression.

- a** $x + x + x + x$ **b** $u \div 5$ **c** $2y - y$
d $5 \times a \times 2$ **e** $8 \times c \times b \times 3 \times a$ **f** $2 \times m \times w \times 10 \times w$

5.03

5 Write each of these in expanded form.

- a** $4mnk$ **b** $\frac{6a}{b}$ **c** $9c^2$

5.04

6 Write an algebraic expression for each statement. Use N to represent any number.

- a** 3 times the number **b** the difference between the number and 5
c the next consecutive number **d** one-third of the number

5.04

7 Write an expression for:

- a** the sum of M and 3 **b** 5 more than B **c** $2H$ decreased by k

5.05

8 Find the value of these expressions if $a = 2$, $b = 5$ and $c = 6$.

- a** $a + b$ **b** $3a - c$ **c** $a + 2b - c$ **d** abc

5.05

9 If $C = 30h + 19$, find C when

- a** $h = 3$ **b** $h = -1$ **c** $h = 5$ **d** $h = 2.5$

5.07

10 Simplify each expression by collecting like terms.

- a** $3m + 4m$ **b** $6uv + 4 + 12uv$ **c** $5h^2 + 6h + 8h^2 + 11h$
d $12m - 4m$ **e** $7y - 4y - y$ **f** $13h^3 - 6h - 8h - 11h^3$

5.08

11 Simplify each expression.

- a** $3m \times 4m$ **b** $9v \times 4 \times 5v$ **c** $3pq \times 5pq$
d $a \times a \times a \times a$ **e** $-7d \times (-3d)$ **f** $lmn \times lmn \times lmn$

12 Simplify each expression.

a $15m \div 3$

b $9v \div v$

c $\frac{49y}{7}$

d $\frac{12x^2y}{4xy}$

e $32de^2 \div 10de$

f $\frac{-3x^2y \times 2x}{3x \times (-x)}$

5.09

13 Solve each equation using the 'guess, check and improve' method.

a $4m - 9 = 7$

b $\frac{k+2}{7} = 2$

c $\frac{d}{3} - 4 = 2$

5.10

14 Solve each equation algebraically, showing all steps.

a $w + 17 = 52$

b $e - 12 = 0$

c $5p = 15$

d $\frac{x}{2} = 6$

5.11

15 Solve each equation algebraically, showing all steps.

a $3t + 4 = 10$

b $2n + 6 = 12$

c $4y - 7 = 5$

d $\frac{m}{4} - 8 = 9$

e $\frac{x}{3} + 5 = 8$

f $\frac{m-4}{2} = 3$

5.12

16 a When a certain number is multiplied by 17, the product is 1003. Which equation can be used to solve this problem? (Let N represent the number.)

Select the correct answer **A**, **B**, **C** or **D**.

A $N + 17 = 1003$

B $1003 - N = 17$

C $17N = 1003$

D $1003 + N = 17$

5.13

b Solve the equation to solve the problem.

17 Lara organises a charity raffle at her school. The only cost is \$180 for prizes.

5.13

a If she sets the price of raffle tickets at \$5, write an equation to find how many tickets she must sell to just cover costs, and solve the equation. (Let N represent the number of tickets.)

b To make a profit of \$2100, how many tickets must she sell?



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